KARTHIK S

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Summary

Artificial Intelligence and Machine Learning undergraduate with hands-on experience in Python, FastAPI, REST APIs, and AI application development. Skilled in building AI-powered applications using Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG). Strong understanding of core Generative AI concepts, including prompt engineering, embeddings, vector databases, transformers, attention mechanisms, fine-tuning, LLM inference, and AI agent architectures. Familiar with LangChain, LangGraph, MCP, Git, and Docker. Strong problem-solving foundation through consistent DSA practice, with a passion for developing practical, scalable, and production-ready AI solutions.

Technical Skills

Programming Languages: Python, JavaScript
Generative AI: LLMs, RAG, Prompt Engineering, Embeddings, Vector Databases, Fine-Tuning, LangChain, LangGraph, MCP, AI Agents
Machine Learning: Supervised Learning, Unsupervised Learning, Feature Engineering, Model Evaluation
Deep Learning: Neural Networks, Backpropagation, Optimization, LSTM, GRU, Transformers
Frontend Development: HTML, CSS, JavaScript, React.js
Backend Development: FastAPI, REST APIs
Databases: PostgreSQL, ChromaDB, Milvus
Tools & Platforms: Git, GitHub, Docker, VS Code, Hugging Face, Google Colab, Ollama, n8n

Education

Sri Shakthi Institute of Engineering and Technology, Coimbatore 2023 – 2027
B.Tech in Artificial Intelligence & Machine Learning

Kshatriya Vidhyasala Higher Secondary School, Virudhunagar 2021 – 2023
Higher Secondary Certificate (HSC)

Projects

TRAVELMATE AI Python, FastAPI, LangGraph, Llama 3.2, Ollama, MCP, Google Maps API, OpenWeather API

- Built a travel planning platform that generates personalized day-by-day itineraries using real-time weather, route, traffic, and place information.
- Implemented an MCP-based tool architecture with Weather, Route, and Places tools, enabling the LangGraph agent to access live external travel data and provide updated travel recommendations.
- Developed a React-based interactive travel interface with dynamic route updates, traffic/route disruption alerts, and alternative route suggestions where available.

AGENT WAF Python, FastAPI, LangGraph, LangChain, Groq, Nginx

- Built a security gateway that acts as a transparent layer between AI agents and external tools, protecting tool invocations from malicious or unsafe requests.
- Implemented a security engine with prompt-injection detection, rate limiting, parameter validation, and tool access control, blocking suspicious tool calls and logging security events.
- Developed a real-time security dashboard to monitor security events, with LangGraph-based agent workflows supporting external tools such as Currency API, along with PDF-based knowledge interaction.

AI SWEETS SHOP React, Python, PostgreSQL, RAG, ChromaDB, Llama 3.2, Ollama

- Built a full-stack sweets-shop website with an integrated AI assistant, allowing customers to browse products and get information about prices, availability, offers, delivery, and shop timings.
- Implemented RAG with ChromaDB to retrieve relevant knowledge using semantic search, while PostgreSQL handles structured product and business data for accurate responses.
- Integrated a locally hosted Llama 3.2 model through Ollama to generate responses using retrieved product and business information, reducing dependency on external LLM APIs.

Certifications & Achievements

- Lean Six Sigma Yellow Belt – Vision Excellence Consulting Pvt. Ltd., Chennai
- Finalist – Flinders University AI Competition 2026, Flinders University, Australia